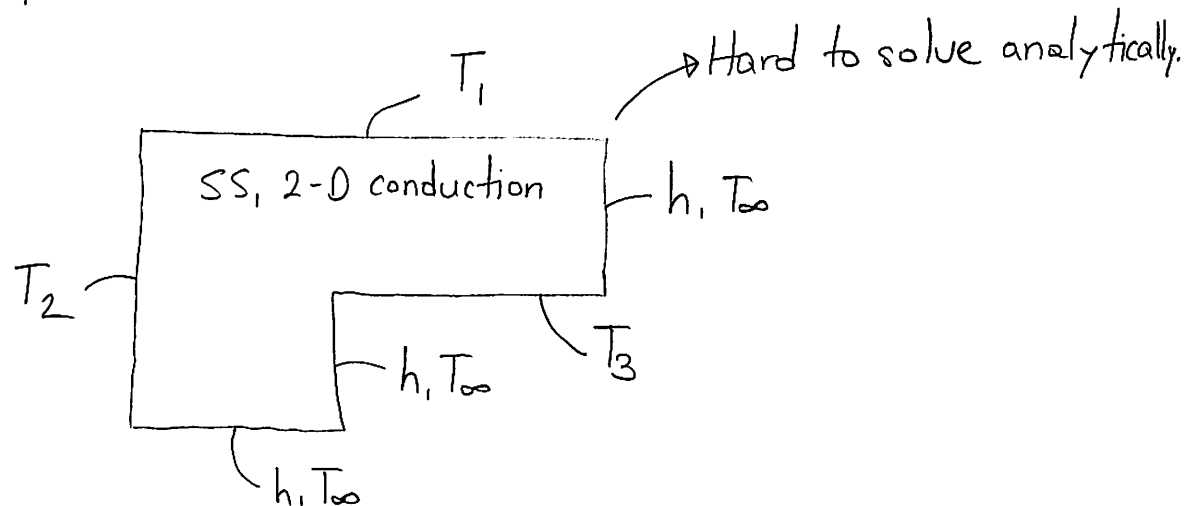


Numerical Analysis of Heat Conduction

So far, we have been using strictly analytical methods to solve heat transfer problems.

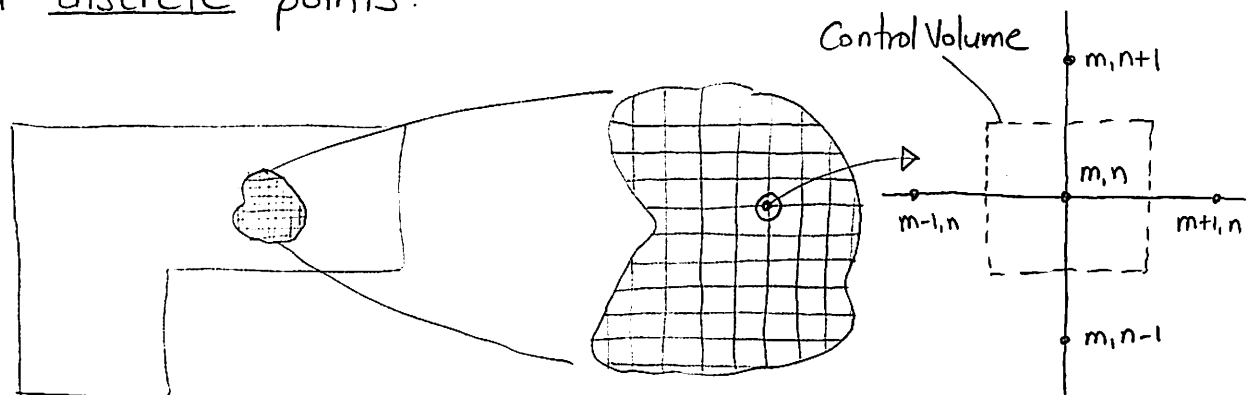
Sometimes, we are presented with non-trivial geometries which require numerical solution of the heat equation.

For example:



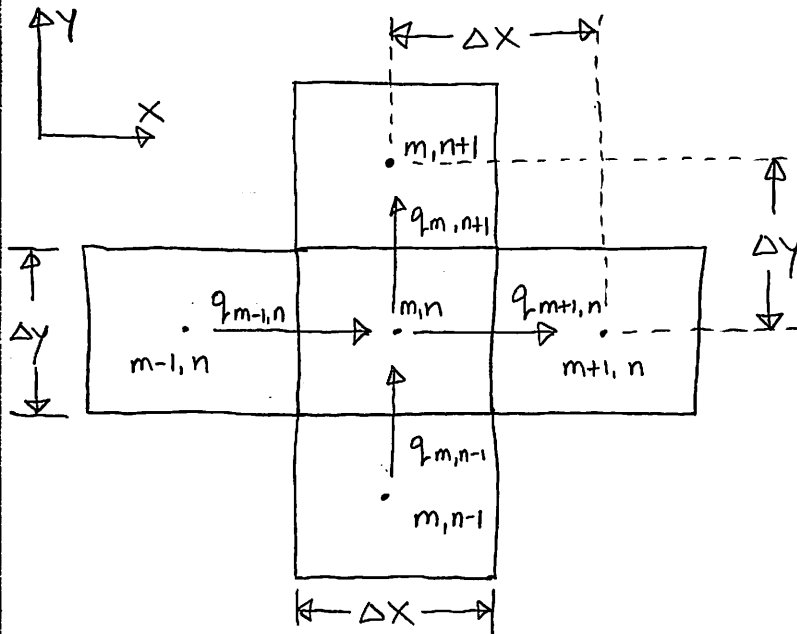
So how do we solve numerically:

First we need to break up our domain into a finite set of discrete points:



Assume $T_{n,m}$ is the average temperature of the control volume. (86)

Applying an energy balance on our control volume:



$$\dot{E}_{in} - \dot{E}_{out} = 0 \quad (\text{Assuming SS, unit depth (2D), } \dot{Q}''' = 0)$$

Applying Fourier's Law:

$$q = -kA \frac{\partial T}{\partial x}$$

$$q_{m-1, n \rightarrow m, n} = k \Delta y \frac{T_{m-1, n} - T_{m, n}}{\Delta x}$$

↙ Note, negative sign dropped because I flipped ΔT .

$$q_{m, n \rightarrow m+1, n} = k \Delta y \frac{T_{m, n} - T_{m+1, n}}{\Delta x}$$

We can do the same for the y-direction & substitute into our energy balance:

$$\begin{aligned} & k \frac{\Delta y}{\Delta x} (T_{m-1, n} - T_{m, n}) - k \frac{\Delta y}{\Delta x} (T_{m, n} - T_{m+1, n}) \\ & + k \frac{\Delta x}{\Delta y} (T_{m, n-1} - T_{m, n}) - k \frac{\Delta x}{\Delta y} (T_{m, n} - T_{m, n+1}) = 0 \end{aligned}$$

Assuming $\Delta x = \Delta y$, we obtain:

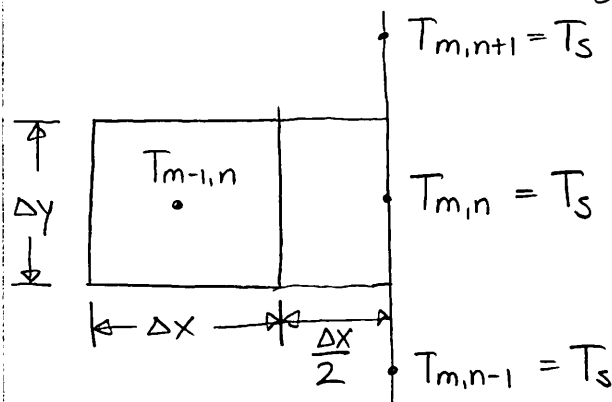
$$\frac{T_{m-1,n} + T_{m+1,n} + T_{m,n-1} + T_{m,n+1}}{4} = T_{m,n}$$

⇒ The nodal temp. is the average of its neighbours.

We would apply this equation for all interior nodes.

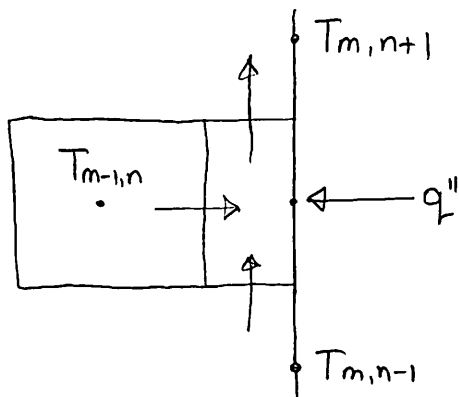
Boundary Conditions (Applied to all surface or boundary nodes)

- ① Fixed Temperature - set node temperature to equal the boundary condition.



For all other B.C.'s see Table 4.2, pg. 246 of Textbook.

- ② Uniform Heat Flux (q'')



Applying an energy balance:

$$\dot{E}_{in} - \dot{E}_{out} = 0$$

$$k \frac{\Delta y}{\Delta x} (T_{m-1,n} - T_{m,n}) + \frac{k \Delta x}{2 \Delta y} (T_{m,n-1} - T_{m,n}) - k \frac{\Delta x}{2 \Delta y} (T_{m,n} - T_{m,n+1}) + q'' \Delta y = 0$$

If $\Delta x = \Delta y$:

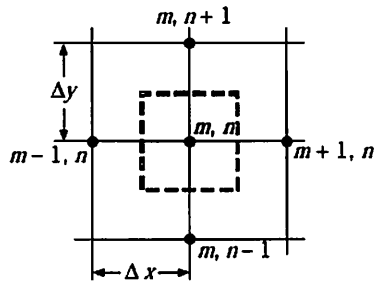
$$2T_{m-1,n} + T_{m,n-1} + T_{m,n+1} + \frac{2q''\Delta x}{k} - 4T_{m,n} = 0$$

Special case ($q'' = 0$): $T_{m,n} = \frac{1}{4} (2T_{m-1,n} + T_{m,n-1} + T_{m,n+1})$ (88)

Table 4.2 of Incropera

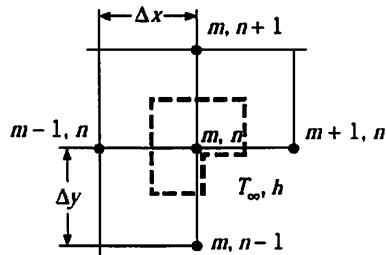
Configuration

Finite-Difference Equation for m, n



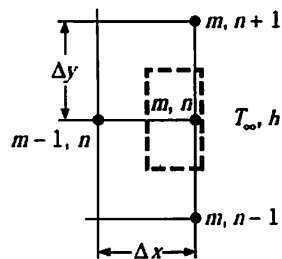
$$T_{m,n+1} + T_{m,n-1} + T_{m+1,n} + T_{m-1,n} - 4T_{m,n} = 0$$

Case 1. Interior node



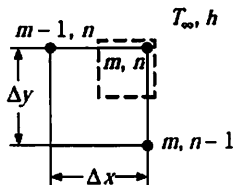
$$2(T_{m-1,n} + T_{m,n+1}) + (T_{m+1,n} + T_{m,n-1}) + 2\frac{h\Delta x}{k}T_{\infty} - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$$

Case 2. Node at an internal corner with convection



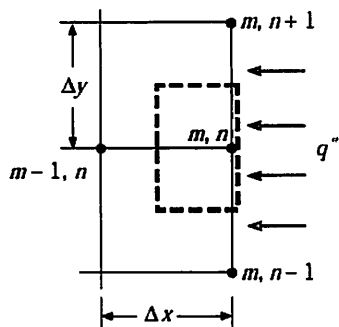
$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2h\Delta x}{k}T_{\infty} - 2\left(\frac{h\Delta x}{k} + 2\right)T_{m,n} = 0$$

Case 3. Node at a plane surface with convection



$$(T_{m,n-1} + T_{m-1,n}) + 2\frac{h\Delta x}{k}T_{\infty} - 2\left(\frac{h\Delta x}{k} + 1\right)T_{m,n} = 0$$

Case 4. Node at an external corner with convection



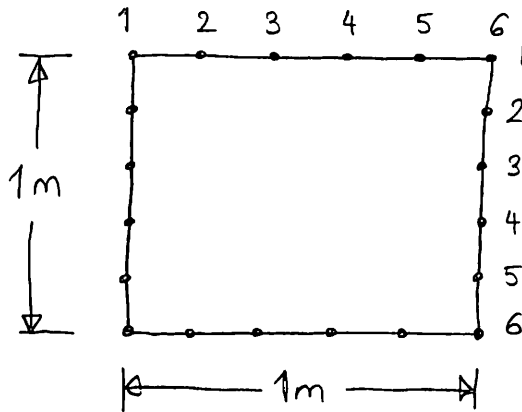
$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2q''\Delta x}{k} - 4T_{m,n} = 0$$

Case 5. Node at a plane surface with uniform heat flux

^{a,b} To obtain the finite-difference equation for an adiabatic surface (or surface of symmetry), simply set h or q'' equal to zero.

Solution Methodology

① Discretize the solution domain



$$m = 6, n = 6$$

$M = \#$ of nodes in x-direction
 $N = \#$ of nodes in y-direction

$$\Delta x = \frac{1}{M-1} = 0.2 \text{ m}$$

$$\Delta y = \frac{1}{N-1} = 0.2 \text{ m}$$

$\left. \begin{array}{l} \# \text{ of interior nodes} = 16 \\ \# \text{ of exterior nodes} = 20 \end{array} \right\} 36 \text{ total nodes}$

② Form finite volume equations for all internal nodes

$$2 \leq m \leq M-1, \quad 2 \leq n \leq N-1$$

③ Form finite volume equations for all boundary nodes

④ Solve $M \times N$ equations and unknowns using:

- i) Gaussian elimination (direct)
- ii) Gauss-Seidel method (indirect and iterative)
- iii) Software (i.e. Excel, Matlab, Maple, etc...)



For the convective boundary condition, look at Table 4.2, eq. 4.42^a

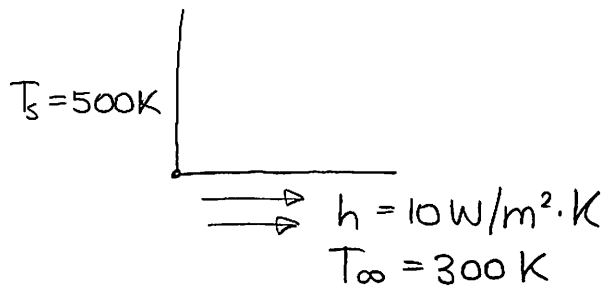
$$(2T_{m,n+1} + T_{m-1,n} + T_{m+1,n}) + \frac{2h\Delta y}{k}T_{\infty} - 2\left(\frac{h\Delta y}{k} + 2\right)T_{m,n} = 0$$

Back substituting h & T_{∞} , & solving for $T_{m,n}$:

$$T_{m,n} = \frac{1}{9}(2T_{m,n+1} + T_{m-1,n} + T_{m+1,n} + 1500)$$

↳ 3 equations

But what about the corners?



Assume:

$$\begin{matrix} T_{1,1} = T_s \\ T_{5,1} = T_s \end{matrix} \rightarrow 2 \text{ equations}$$

- ④ Assemble the equation set: 25 equations, 25 unknowns
- ⑤ Solve the set of equations (we will use Excel)